INCH-POUND

MIL-M-28787/344
AMENDMENT 2
30 July 1993
SUPERSEDING
AMENDMENT 1
17 July 1984

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

ENCODER, FAULT ISOLATION, KEY CODE MFG

This amendment forms a part of Military Specification MIL-M-28787/344(EC), dated 18 April 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

*TABLE I: Delete and substitute new table I as printed on pages 3 and 4 of this amendment.

PAGE 5

*TABLE I, Test conditions N1, N2 and N3-N6, Test type column: Delete "X" (two places) and substitute "S" (two places).

PAGE 6

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through P shall be performed in any order with the following exceptions. Test conditions A, L, M, and N are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, L, M, or N are part of a sequence of tests, the order of testing shall be as follows: A, L, M, and N where applicable, followed by the remaining tests in the sequence."

*4.5.1, fifth line: Delete "B through P" and substitute "B through K and P".

4.5.2, fourth line: Add "and test condition J" between the words "column" and "shall".

*4.5.3, third line: Delete "B through P" and substitute "B through K and P".

*4.5.3, sixth line: Delete "A through P" and substitute "B through K and P".

PAGE 8

FIGURE 2: Delete and substitute new figure 2 as printed on page 5 of this amendment.

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PAGE 9

TABLE II: Delete and substitute new table II as printed on pages 6 and 7 of this amendment.

PAGE 14

TABLE IV, Test conditions D1 and D2, Min column: Delete "2.5" (two places) and substitute "2.50" (two places).

TABLE IV, Test conditions D1 and D2, Max column: Delete "4.5" (two places) and substitute "4.55" (two places).

PAGE 22

***FIGURE 8:** Delete and substitute new figure 8 as printed on page 8 of this amendment.

PAGES 24 AND 25

***FIGURE 9:** Delete and substitute new figure 9 as printed on page 9 of this amendment.

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FIGURE 15, note 4: Delete and substitute the following:

"4. Adjust item P07-2 as follows:
0.7 V ± 1% for test conditions P1-P20 and P25-P44.
2.5 V ± 1% for test conditions P21-P24 and P45-P48."

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FIGURE 18, note 2, first line: Delete "16" and substitute "11.8".

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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(Project 5963-0259)

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TABLE I. Module electrical reequirements.

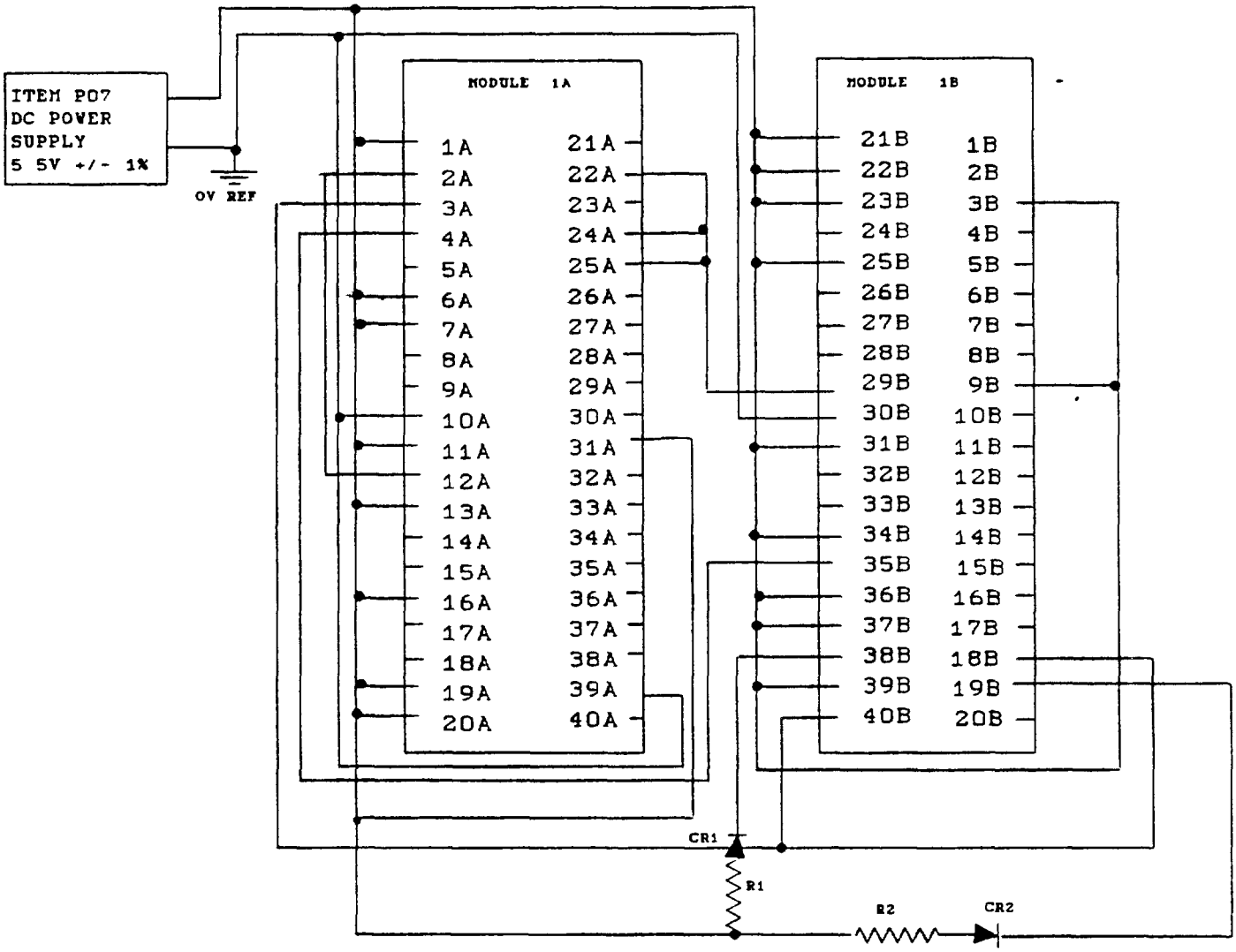
Test cond	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ± 5°C				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
A3-A16	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	Resistance	III	X	2024	2376	1914	2574	2024	2376	1914	2574	Ω
C1,C2	Resistance	III	X	138.0	162.0	130.5	169.5	138.0	162.0	130.5	169.5	Ω
D1,D2	NOR gate operation	IV	X	2.50	4.55	2.50	4.55	2.50	4.55	2.50	4.55	V
D5,D6, D10,D11	NOR gate operation	IV	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
D3,D4, D7-D9, D12	NOR gate operation	IV	X	0	400	0	400	0	400	0	400	mV
E1,E11	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	0	400	0	400	0	400	0	400	mV
E2-E10, E12-E19	Exclusive OR/8 INPUT NAND gate operation (module 1A)	V	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
F1,F11	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	0	400	0	400	0	400	0	400	mV
F2-F10, F12-F19	Exclusive OR/8 INPUT NAND gate operation (module 1B)	VI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G1-G3, G5-G7	Independent NAND gate operation	VII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
G4,G8	Independent NAND gate operation	VII	X	0	400	0	400	0	400	0	400	mV

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TABLE I. Module electrical reequirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ± 5°C				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
H1-H5, H11-H13, H18	Encoder fault isolation	VIII	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
H6-H10, H14-H17	Encoder fault isolation	VIII	X	0	400	0	400	0	400	0	400	mV
J	Filter capacitance	IX	S	8	12	7	13	7	13	6	14	µF
K	Power supply current (Vcc max)	X	X	10	80	10	80	10	80	10	80	mA

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NOTES:
1. R1 = R2 = 137Ω ± 1%, 1/8W.
2. CR1 = CR2 = 1N4150, or equivalent.

FIGURE 2. Operating temperature, component temperature, and life test circuit.

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TABLE II. Isolation test.

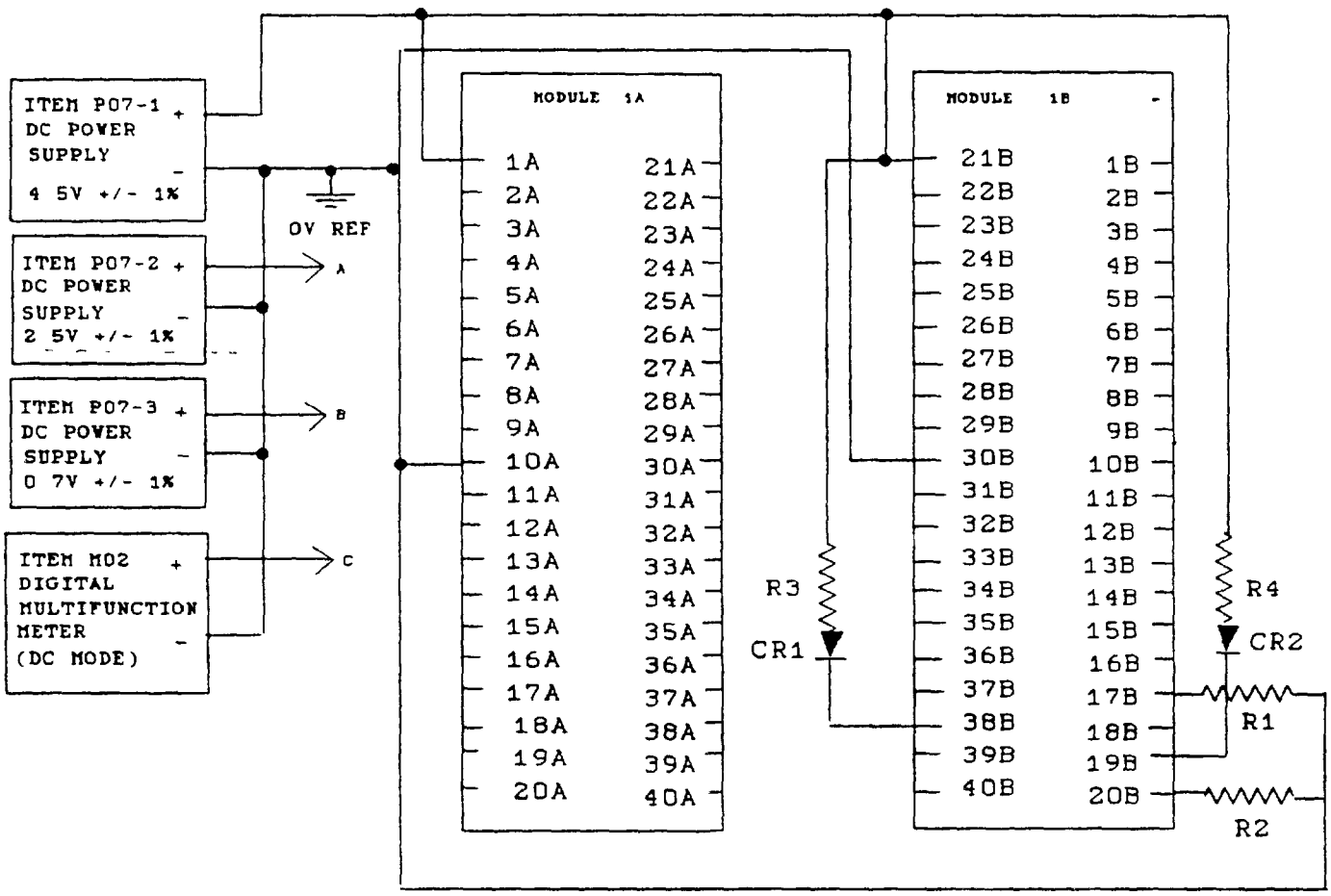
Test cond	Parameter	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	Frame	All pins	10	∞	MΩ	3
A2	Isolation, used to unused pins	1A-7A,10A-13A, 15A,16A,19A, 20A,22A-25A, 31A,37A,39A, 40A,1B,3B,9B, 10B,17B-23B, 25B,29B-31B, 34B-40B	8A,9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A,2B,4B-8B, 11B-16B,24B, 26B-28B,32B, 33B	10	∞	MΩ	3
A3	Isolation, unused pins	8A	9A,14A,17A, 18A,21A,26A-30A,32A-36A, 38A	10	∞	MΩ	4
A4		9A	14A,17A,18A, 21A,26A-30A, 32A-36A,38A				
A5		14A	17A,18A,21A, 26A-30A,32A-36A,38A				
A6		17A	18A,21A,26A-30A,32A-36A, 38A				
A7		18A	21A,26A-30A, 32A-36A,38A				
A8		21A	26A-30A,32A-36A,38A				
A9	Isolation, unused pins	26A-30A	32A-36A,38A	10	∞	MΩ	4

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TABLE II. Isolation test - Continued.

Test cond	Parameter	Connect		Initial limits 25°C ± 5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
A10	Isolation, unused pins	32A-36A	38A	10	∞	MΩ	4
A11	"	2B	4B-8B, 11B-16B, 24B, 26B-28B, 32B, 33B	"	"	"	"
A12	Isolation, unused pins	4B-8B	11B-16B, 24B, 26B-28B, 32B, 33B	10	∞	MΩ	4

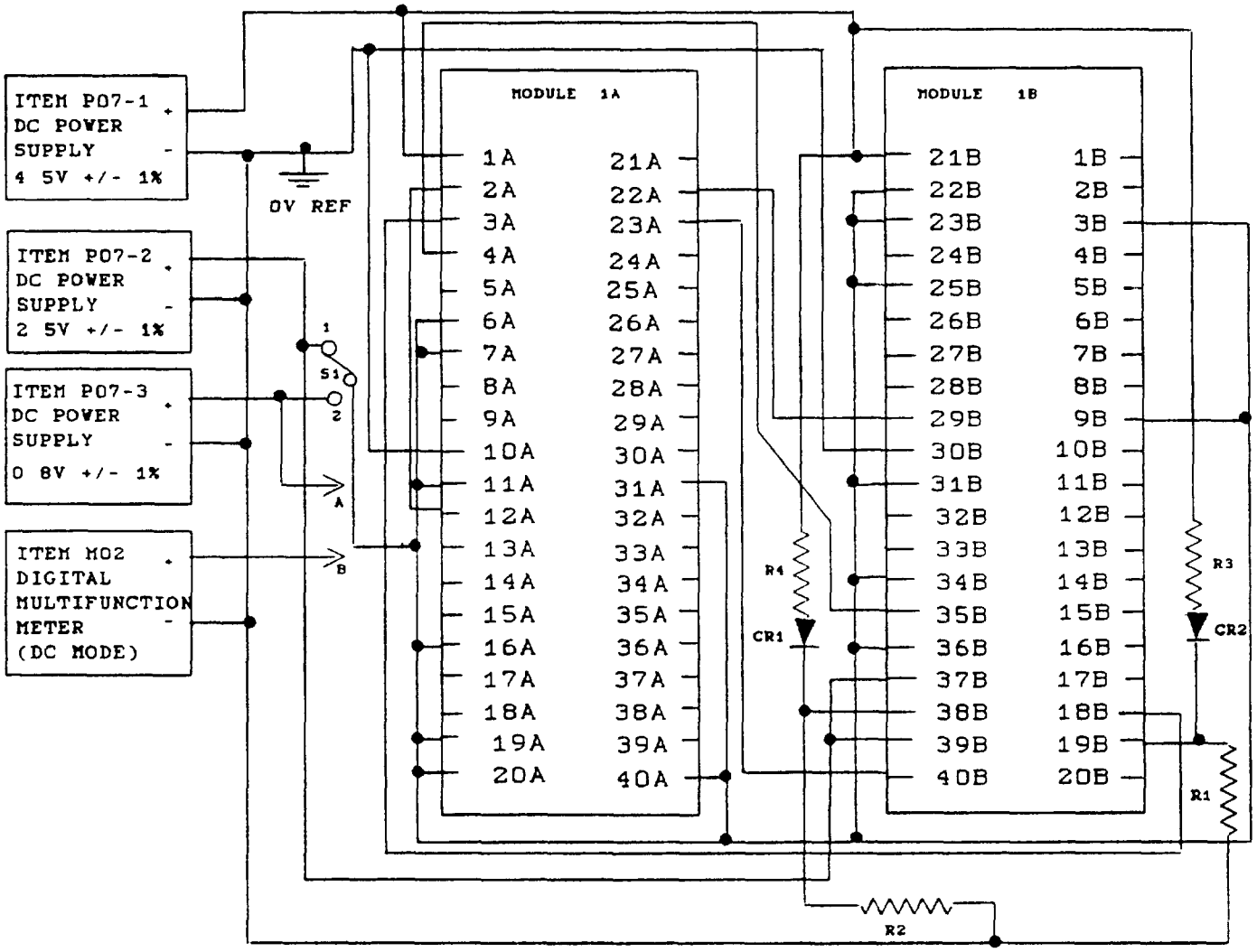
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NOTES:
1. R1 = R2 = 2.08 kΩ ± 1%, 1/8W.
2. R3 = R4 = 97.6Ω ± 1%, 1/8W.
3. CR1 = CR2 = 1N4150, or equivalent.

FIGURE 8. Independent NAND gate operation test circuit for table VII.

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NOTES:
1. R1 = R2 = 2.08 kΩ ± 1%, 1/8W.
2. R3 = R4 = 69.8Ω ± 1%, 1/8W.
3. S1 = SPDT switch.
4. CR1 = CR2 = 1N4150, or equivalent.
5. Set switch S1 to position 1 for test conditions H1, H3-H6, and H8-H10 and to position 2 for test conditions H2, H7, and H11-H18.

FIGURE 9. Encoder fault isolation test circuit for table VIII.

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